

EEL 3111C Lab 3 LTSpice Tips

For 3.4.1 DC operation point simulation is “.op”, not “tran”

You can get it from selecting the “DC op pnt” tab. Don’t use the Transient tab that pop up on the screen.

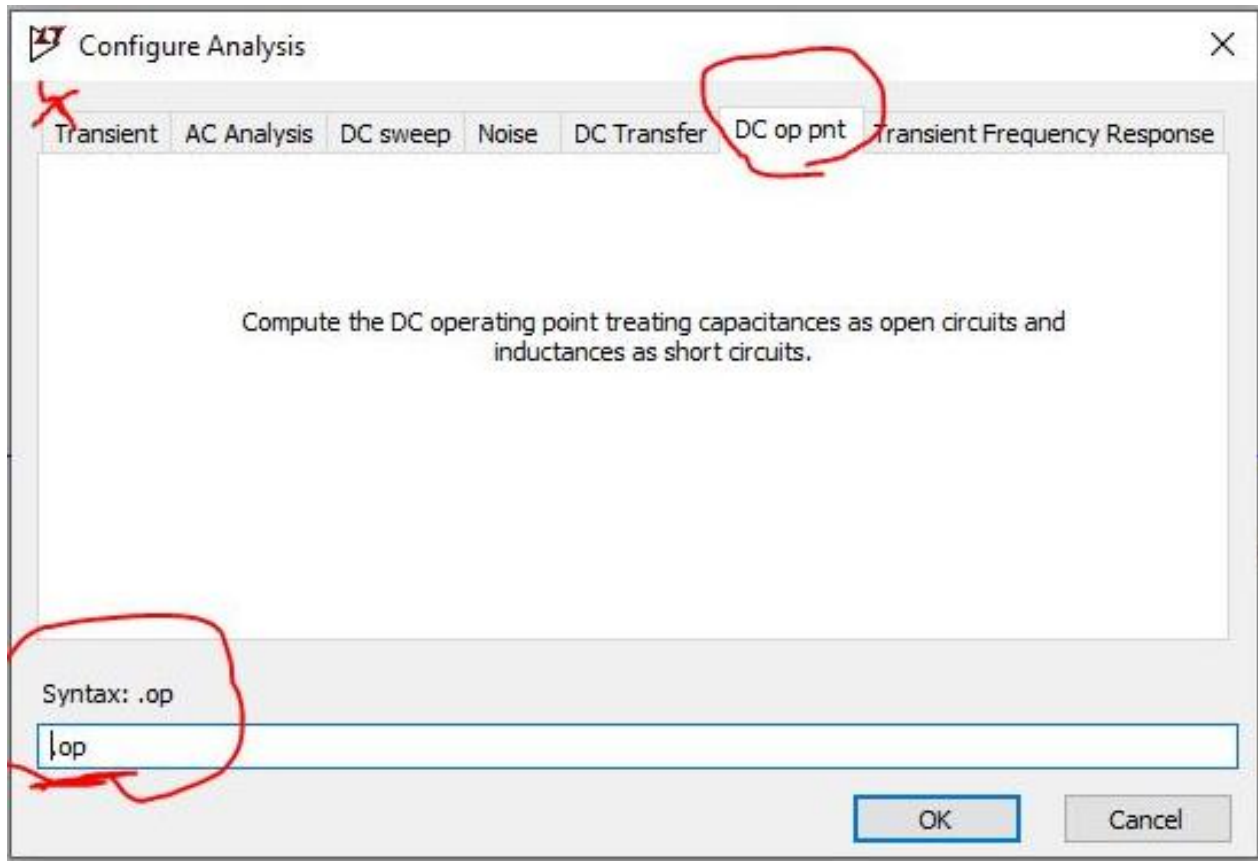


Figure 1 DC operation point simulation tab screenshot

2. Use a voltage source for +5V (make sure you ground the negative side):

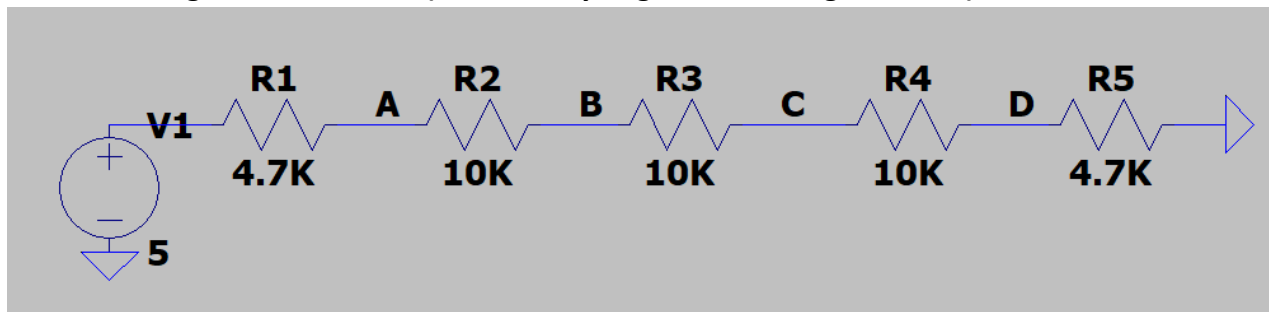


Figure 2 Example of powering up a resistor ladder

3. Use "Label Net" to name the nodes you are going to measure.

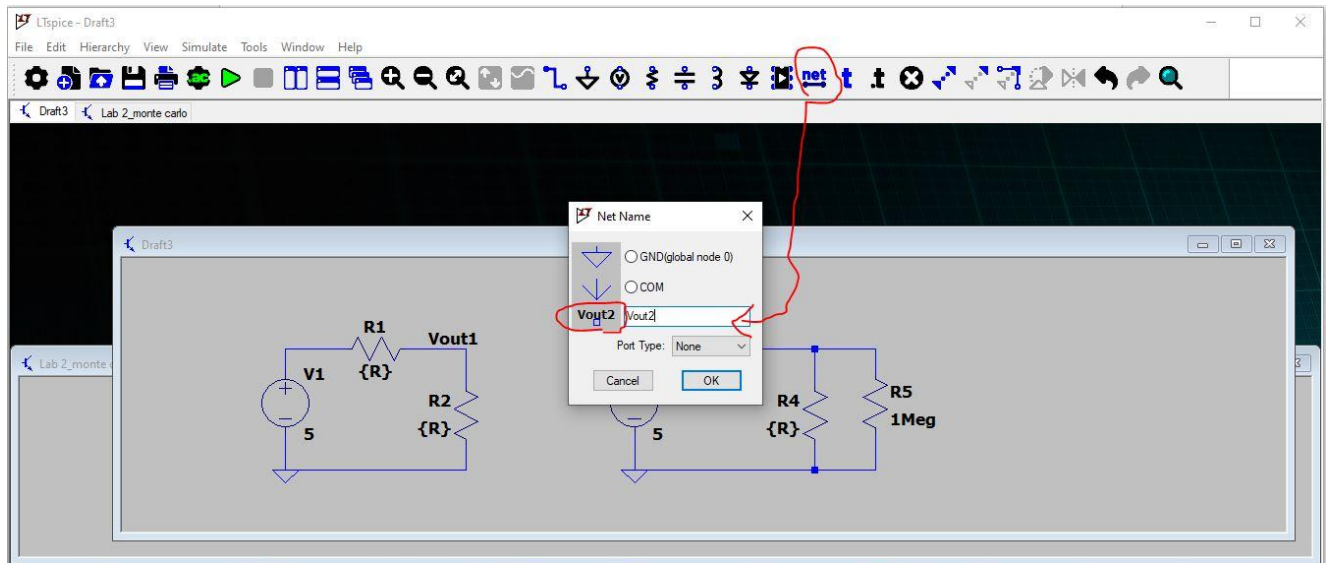
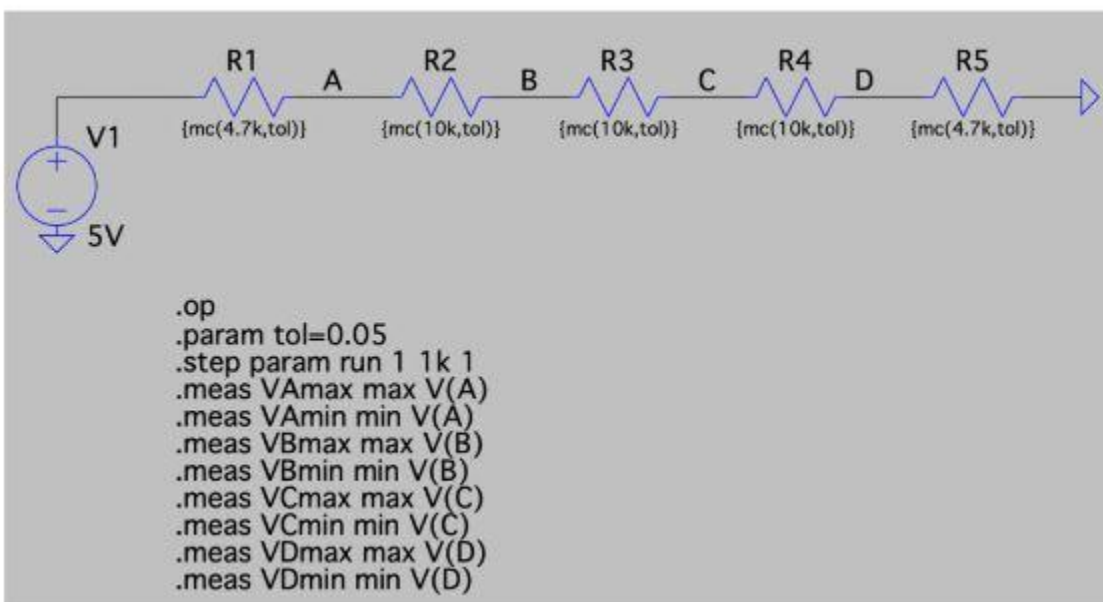


Figure 3 Use "Label Net" to name the nodes in the circuit.

4. Use ".meas" directive to measure voltage ranges of nodes:

Example: **.meas VAmx mas V(A)**

VAmx is the new variable, **max** is the command to find the max of **V(node name)**

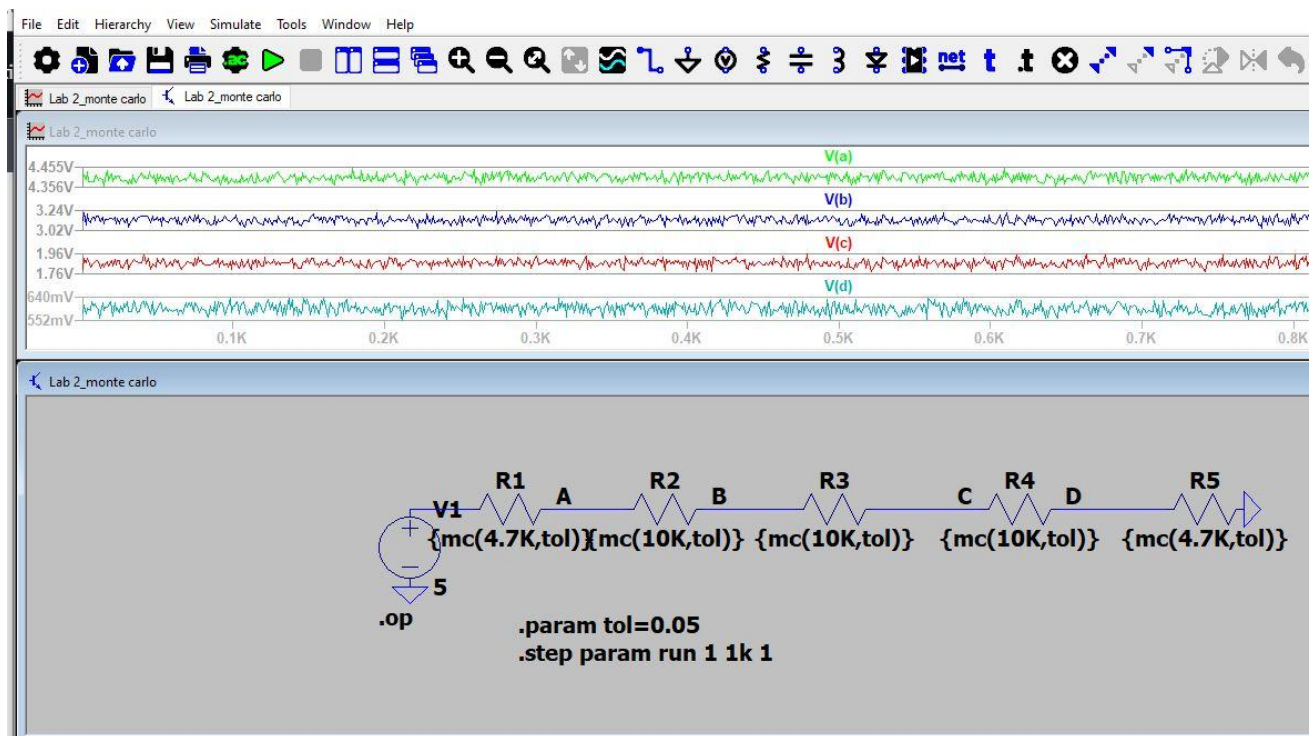


View your measurements by right clicking anywhere on the screen-> View-> Spice output log.

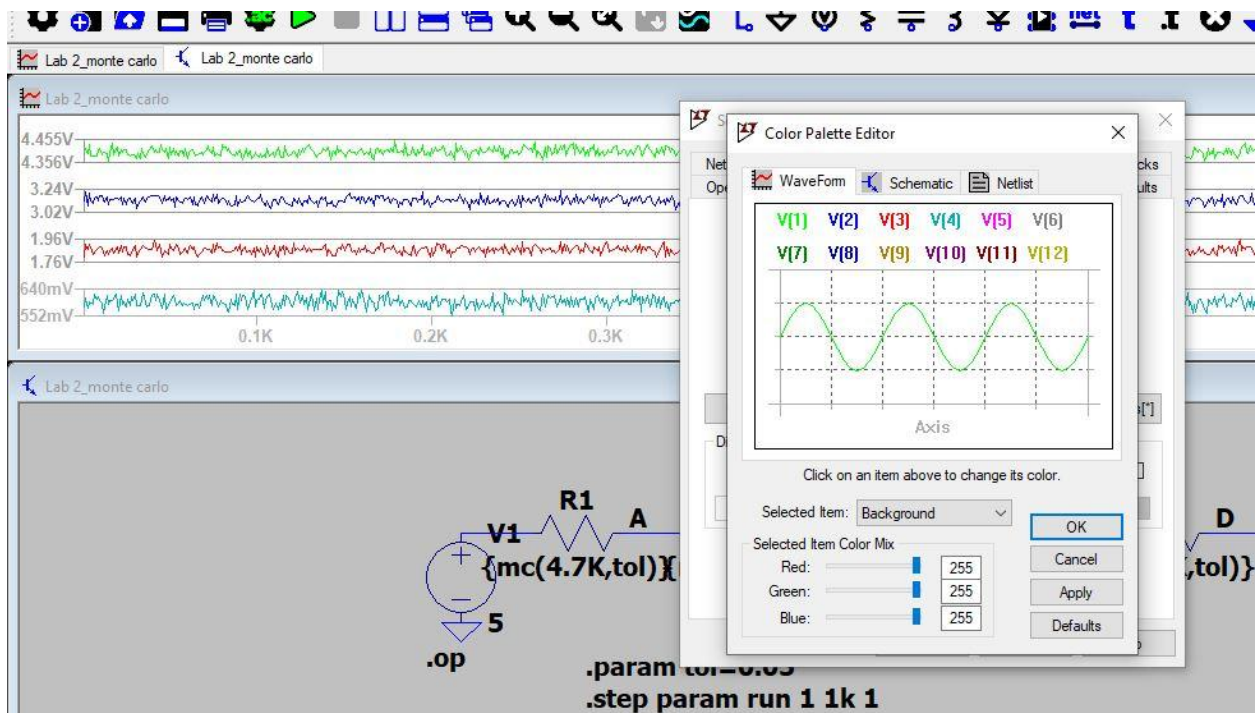
You may have to scroll all the way down, use the sidebar.

However, if your LTSpice “.meas” spice director doesn’t work (maximum and minimum values are the same), you will have to manually find the max and min from the individual plot panes by hovering over the graph and view the Y values.

5. After running your simulation, a waveform window will pop up. To add traces (signals) to the screen, you can hover over the node you want to measure and you will see a red probe. Click on the node and the waveform will appear. You can also right click on the screen to “Add Trace” and “**Add plot pane below**” to probe the next node for another trace to the new plot pane.



6. To change the waveform background color from dark to white (especially for better viewing dark color traces), right click the waveform window and select “**Window settings**”> Color Scheme > Pull down menu, in “select item” scroll down to the **background**, set RGB color numbers all to 255.



7. Copy the simulation results and copy it to word, sometimes the best way is not snipping tools (some distortion).

Use right click => view => copy image to clipboard.